SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I Sanjana.G (192511020) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : Sanjana.G

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1 - AT3 - Cloud Computing Short Answer test

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Reg no: 19251020
Course:
Course code: CSA1520
Slot : A

Questions:

1) Define cloud computing and Explain any four characteristics that make it different from traditional on-premise computing.

2) Trace the Evolution of cloud computing from hardware Evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

3) Explain the role of server virtualization in cloud computing as a service model.

4) Discuss the role of data centers in delivering cloud services.

5) Differentiate IaaS, PaaS, SaaS and XaaS with one-suitable real word example for each model.

Answers

1) Definition:

* Cloud computing is a technology that provides computing resources such as servers, storage, databases, networking, software processing, all required services usually on pay-on-use.
* User can owning it whenever they want.

The four characteristics are:

(i) On-demand self-service
(ii) Broad network Access
(iii) Resource pooling
(iv) Rapid scability.

(i) On-demand self-service:

User can create (or) access computing resources whenever required without depending on a service provider.

(ii) Broad network Access:

Cloud services can be accesd through the internet using devices such as laptops, smartphones and tablets.

(iii) Resource pooling:

The provider shares computing resources among multiple user using virtualization.
→ Resource are allocated to the demand.

(iv) Rapid scalability:

Resources can be quickly increased (or) decreased based on workload.

Ex: online shopping website can increase screen capacity during festival.

2) Trace Evolution of cloud Computing

Cloud Computing Evoluted through several stages of hardware and software development.

Evolution:-

1) Mainframe computing:

In the early stages. Organisations used very large maintaince and mainframe computers.

* Many users are accessed a single powerful Computer through the terminals.

2). Personal Computers:-

computing became decentralized with the introduction of pc's

* Using ~~machines would~~ could perform computing tasks on individual machine.

3) Networking and Internet

The ~~development~~ of Computer networks and the Internet allowed ~~computers~~ computer ~~to~~ communicate and to share resarces to different locations.

4) Client - Server Computing

Application began using separate client and Server systems.

5) Virtualization

Virtualization allowed on physical ~~system~~ servers to run multiple virtual servers.

6) Internet based software:

Software started being delivered through intanet Instead of being installed locally.

7) Modern Cloud Computing

Cloud providers combine virtualization high speed networks, large data centers to provide Iaas, Paas and Saas.

8) Role of server Virtualization in cloud computing

Server Virtulization is a technology that divides one Physical server into multiple independent virtual machines using software called hypervisors.

Role in cloud computing:

* Improves the utilization.
* Provides Isolation between different users.
*

Answers:

Model	Meaning	What user get	Ex?
Iaas	Infrastructure as service	Virtual machine	Amazon EC2
Paas	Platform as a service	Platform/ tool to develop	Google App Engine
Saas	Software as a serive	Ready to use Software	Google docs
Xaas	Anything / Everthing as service	various IT Service delivered	AWS Lauder